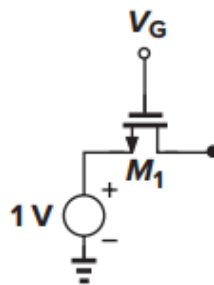


Assignment 1

Subject: VLSI Design, Due date: 06/05/2026

1. Explain the different doping techniques?
2. For $W/L = 50/0.5$, plot the drain current of an NFET and a PFET as a function of $|V_{GS}|$ as $|V_{GS}|$ varies from 0 to 3 V. Assume that $|V_{DS}| = 3$ V.
3. For the arrangement in Fig, plot the on-resistance of M_1 as a function of V_G . Assume that $\mu_n C_{ox} = 50 \mu A/V^2$, $W/L = 10$, and $V_{TH} = 0.3$ V. Note that the drain terminal is open.



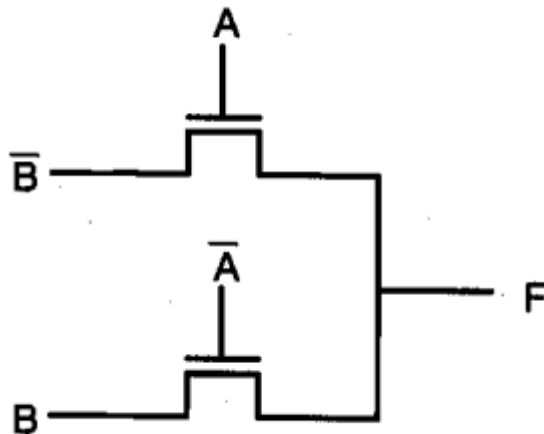
4. Design half adder circuit using pass transistor logic.
5. Explain chemical vapor deposition (CVD) in brief.
6. An NMOS pass transistor has a threshold voltage $V_{th} = 0.7V$ (ignoring body effect) and a supply voltage $V_{DD} = 1.8V$. If the gate and the drain are both connected to V_{DD} , what is the maximum voltage at the source V_{out} ? If we cascade two such NMOS transistors in series, what is the final output voltage?
7. Differentiate between Complementary Pass-Transistor Logic (CPL) and standard PTL. What are the trade-offs regarding routing complexity and speed?
8. Write the all steps involved in photolithography in details.
9. Why the oxidation is needed in fabrication of MOSFET, write the different type of oxidation?
10. Realize $y=AB+C$ using:
 - A) Pass-transistor logic
 - B) Transmission gates
11. Write about the Multi-chip modules.
12. Write short note on Domino logic.

13. What is clocked CMOS dynamic register.
14. Implement negative level triggered D latch using transmission Gate logic
15. Differentiate between pass transistor logic and transmission gate logic.
16. Whether, NMOS transistor pass strong 0 or strong 1 and why?
17. Design and evaluate a circuit to demonstrate the problem of capacitive coupling and clock feed-through in a dynamic logic design.
18. Consider the following silicon junction diode: $C_{j0} = 2 \times 10^{-3} \text{ F/m}^2$, $A_D = 0.5 \text{ } \mu\text{m}^2$, and $\phi_0 = 0.64 \text{ V}$. A reverse bias of -2.5 V . Find junction capacitance.
19. Explain the different mode of operation of MOSFET.
20. Explain the threshold voltage of MOSFET and derive the expression for the same.
21. Write short note on short channel effect in MOSFET.
22. Derive the expression for drain current in velocity saturation mode.
23. Derive the expression of dynamic power in CMOS inverter.
24. What is noise margin.
25. Discuss about the impact of threshold reduction on performance and static power dissipation in CMOS inverter.

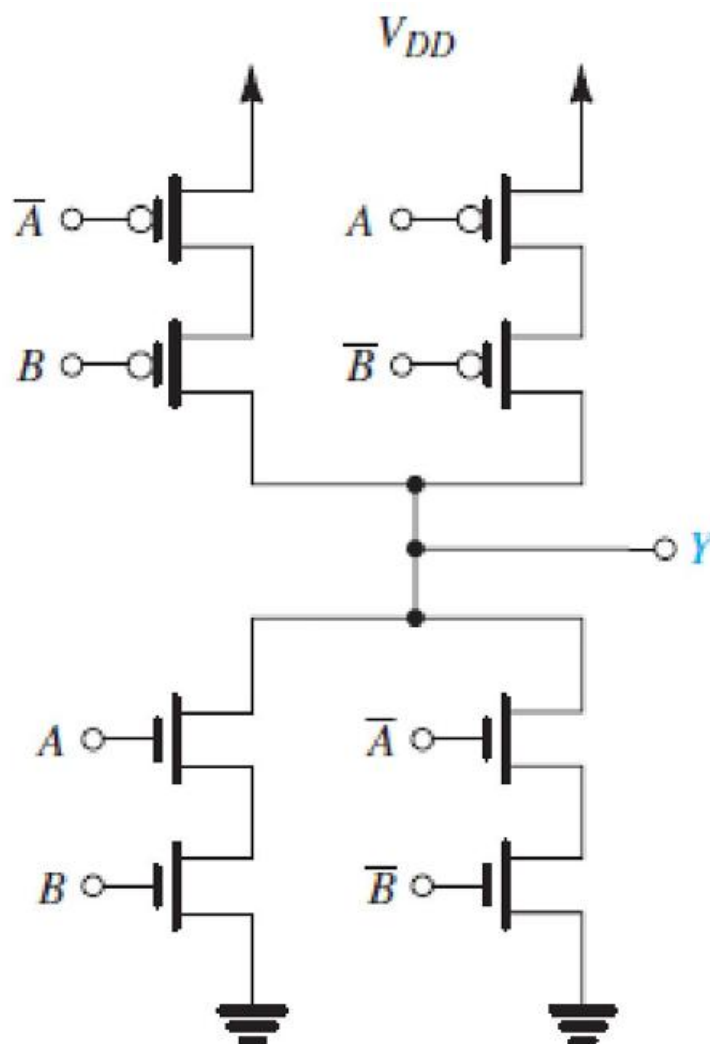
Assignment 2

Subject: VLSI Design, Due date: 08/05/2026

1. Discuss the stacking effect of transistor in CMOS circuit.
2. How does logic reconstructing affect the speed of circuit.
3. Compute the logical effort of four input NOR gate.
4. What is switching activity? What are design techniques to reduce switching activity?
5. Implement two-input multiplexer circuit using two CMOS TGs.
6. Circuit diagram of a CPL-based XOR gate.
7. (a) Determine the logic function F.
(b) Design a circuit to implement the same logic function, but using NOR gates and draw a transistor-level schematic for the function.

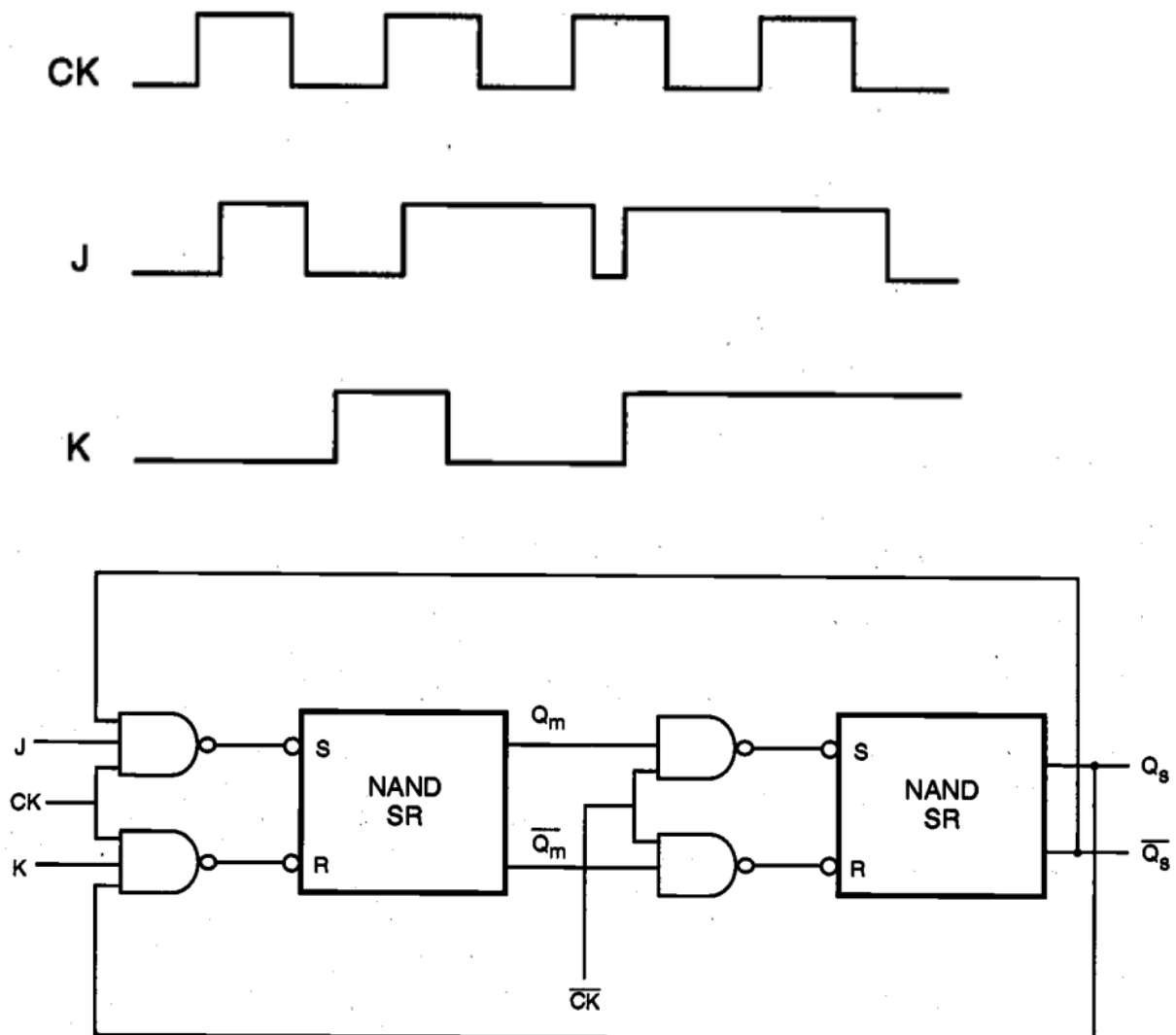


10. For NMOS transistor with $V_t = 0.4 \text{ V}$, $\mu_n C_{ox} = 108 \mu\text{A/V}^2$, $W/L = 10$, $V_{GS} = 1.2 \text{ V}$ and $V_{DS} = 2 \text{ V}$, then find drain current and r_{ds} .
11. What is pinch off voltage, explain in detail.
12. Realise 1-to 4 demux using CMOS.
13. Design a D latch using multiplexer.
14. Find appropriate sizes for the transistors used in the exclusive-OR circuit of given figure. Assume that the basic inverter has $(W/L)_n = 0.20 \mu\text{m}/0.13 \mu\text{m}$ and $(W/L)_p = 0.40 \mu\text{m}/0.13 \mu\text{m}$. What is the total area, including that of the required inverters?



15. Make 8:1 MUX from 2:1 MUX using transmission gate logic.
16. Compare NORA logic and C2MOS logic with example.

17. The voltage waveforms shown below are applied to the nMOS JK master-slave flip-flop shown below. With the flip-flop initially reset, show the resulting waveforms at nodes Q_m (master flip-flop output) and Q_s (slave flip-flop output).



18. Implements the function $F = (XYZ + AB)'$ using dynamic CMOS logic gate.

19. What is the output of circuit given below

